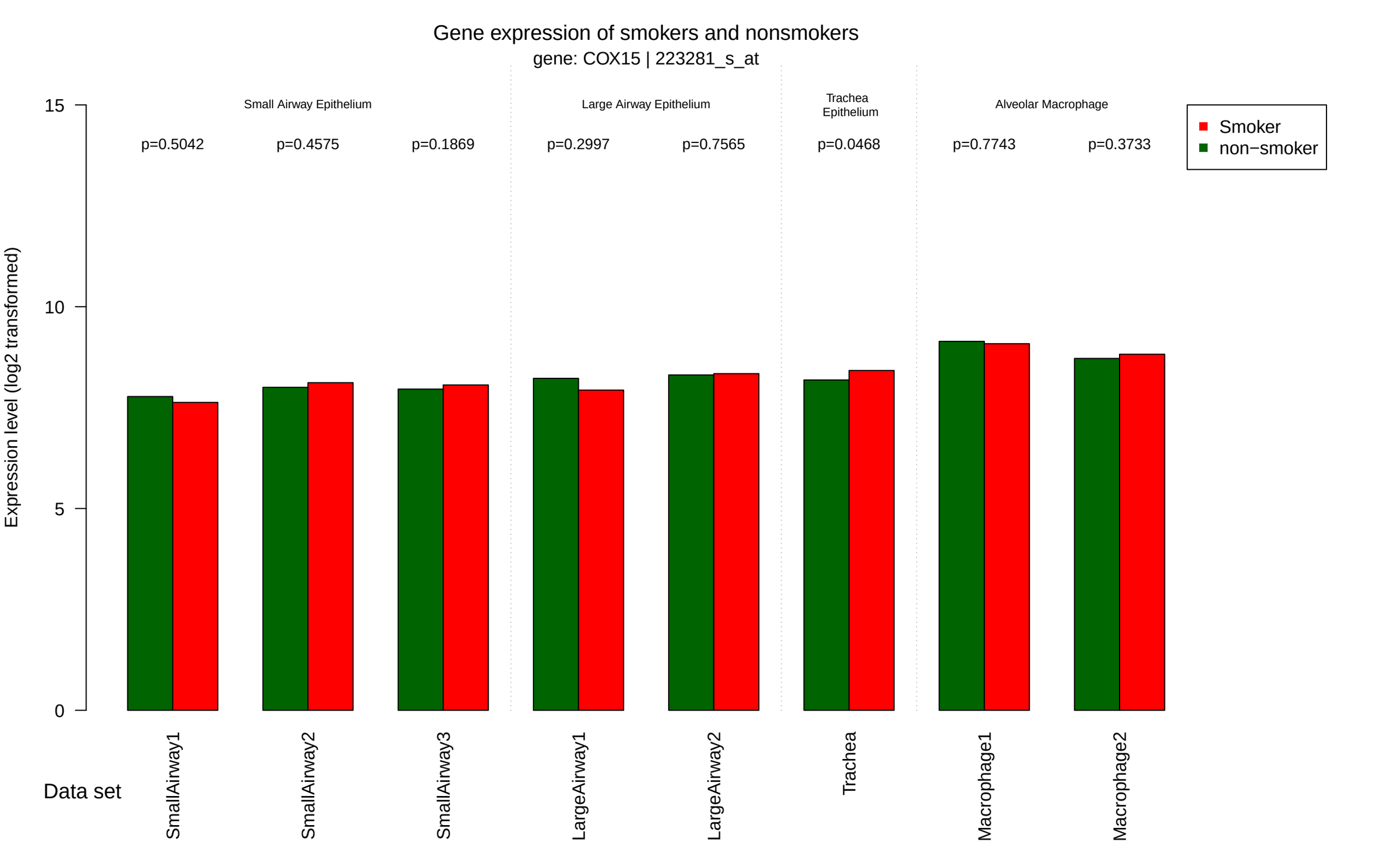
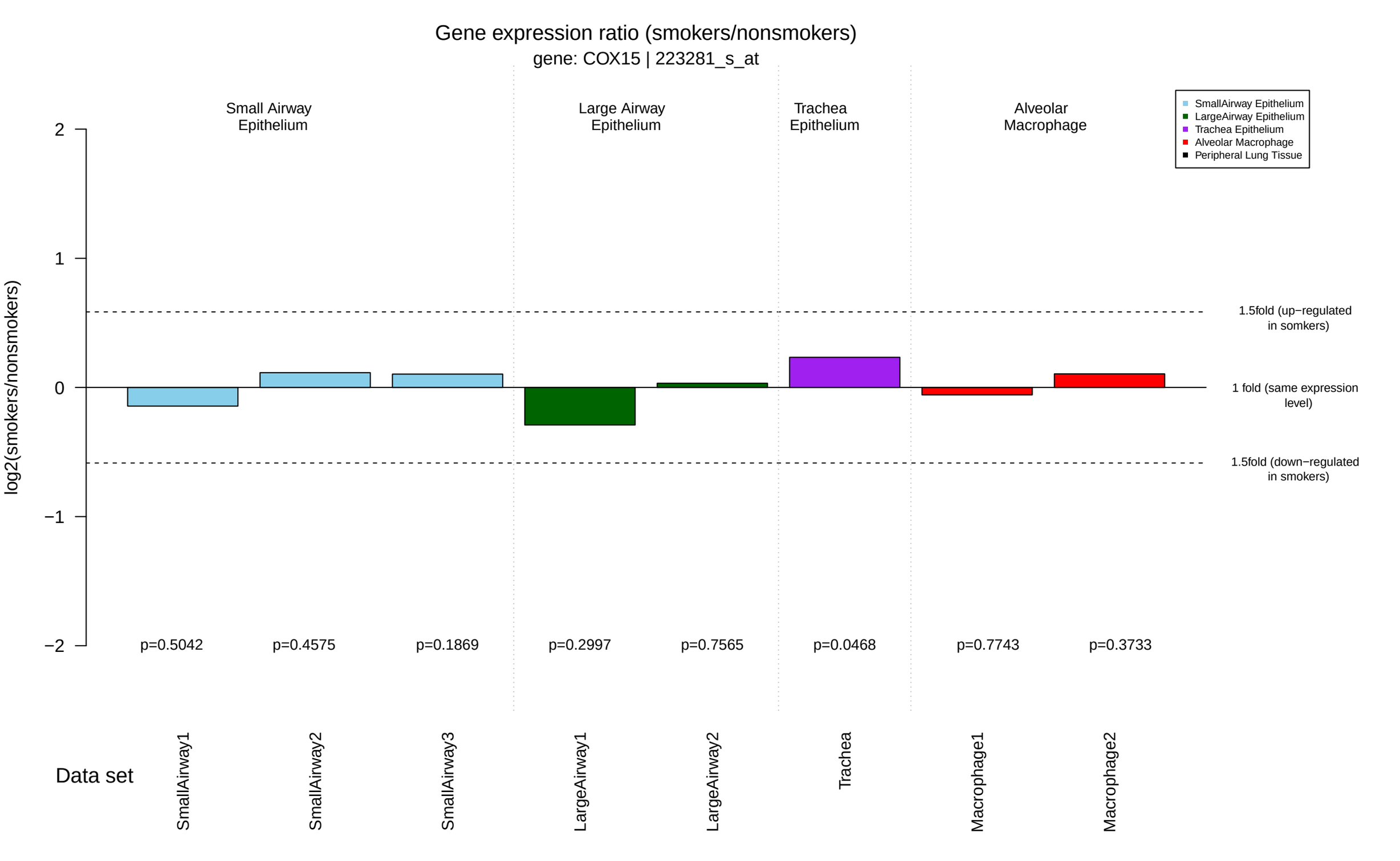
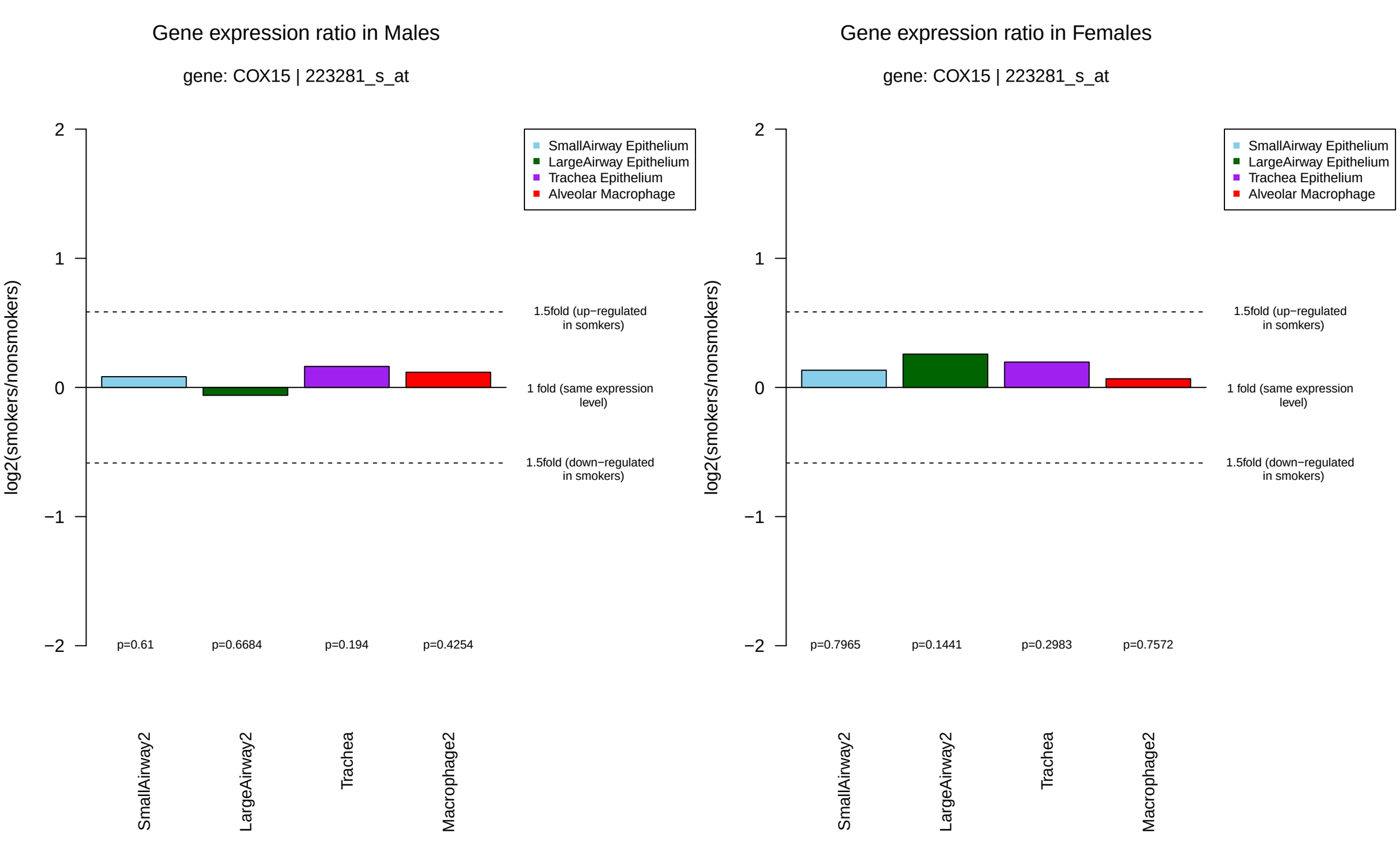
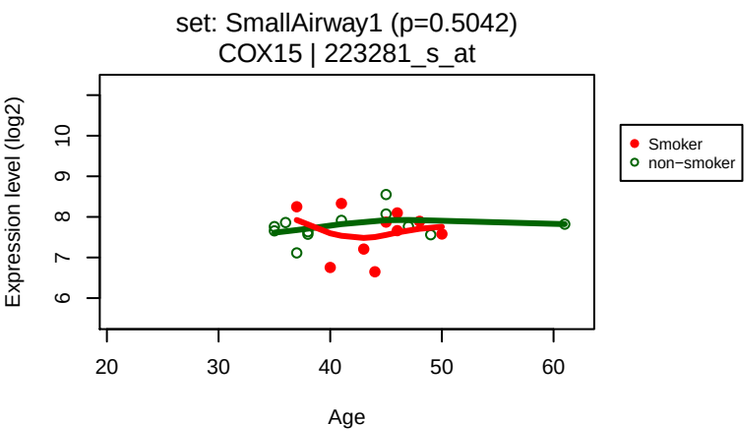
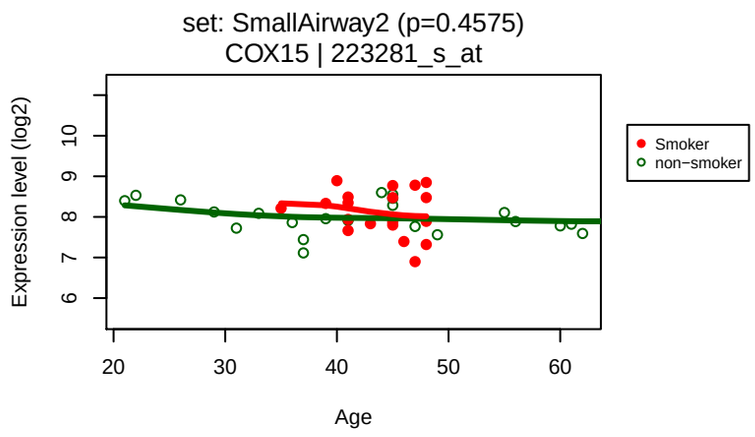
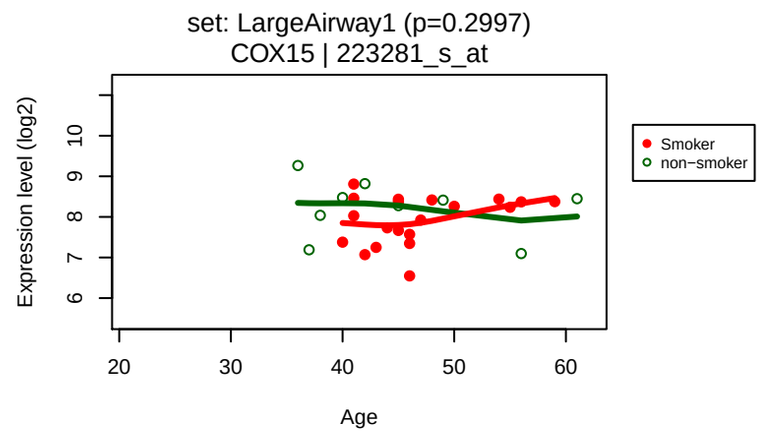

Gene expression of smokers and nonsmokers
gene: COX15 | 223281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5042
p=0.4575
p=0.1869
p=0.2997
p=0.7565
p=0.0468
p=0.7743
p=0.3733
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

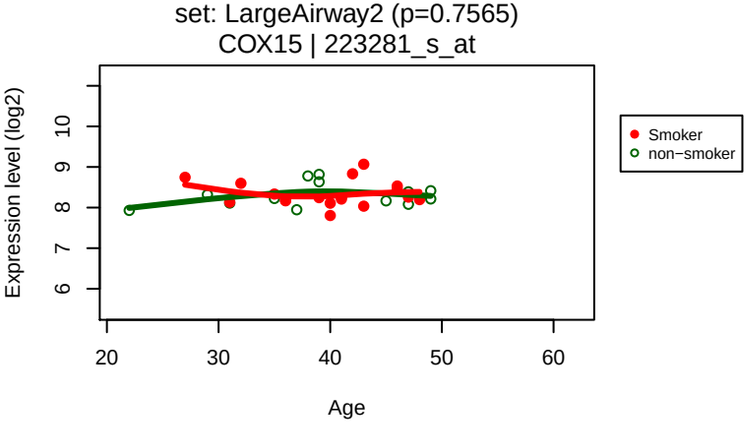

Gene expression ratio (smokers/nonsmokers)
gene: COX15 | 223281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5042
p=0.4575
p=0.1869
p=0.2997
p=0.7565
p=0.0468
p=0.7743
p=0.3733
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

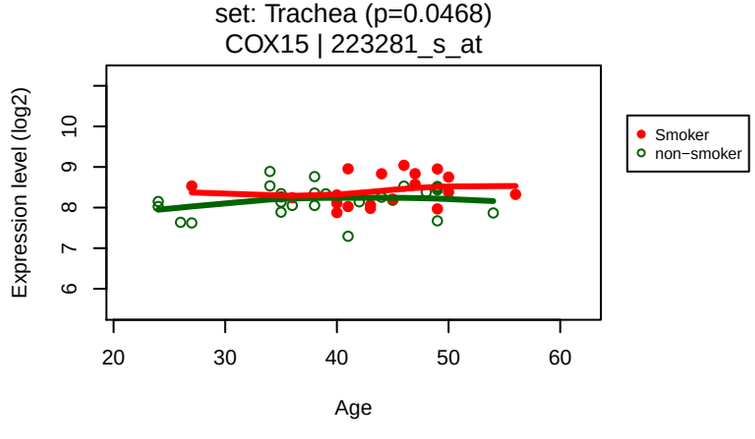

Gene expression ratio in Males
gene: COX15 | 223281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.61
p=0.6684
p=0.194
p=0.4254
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX15 | 223281_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7965
p=0.1441
p=0.2983
p=0.7572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

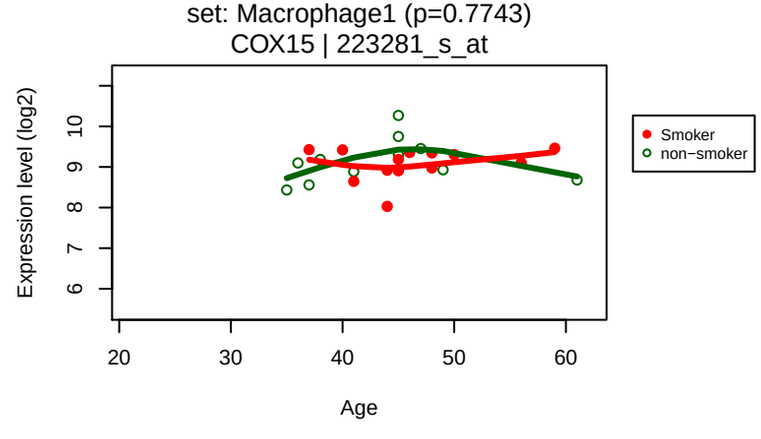

set: SmallAirway1 (p=0.5042)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker

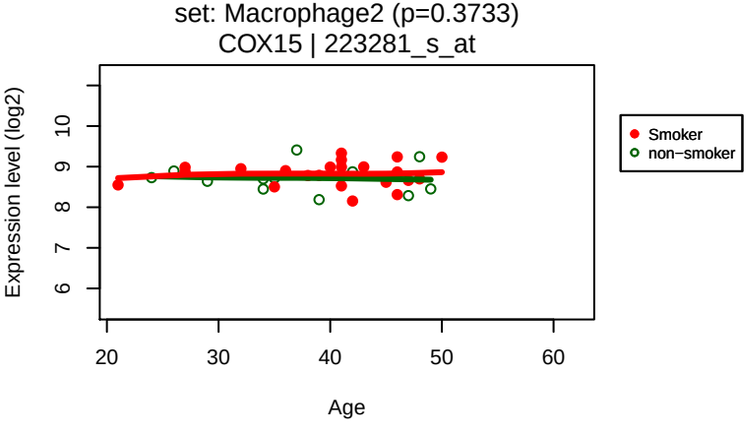

set: SmallAirway2 (p=0.4575)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker

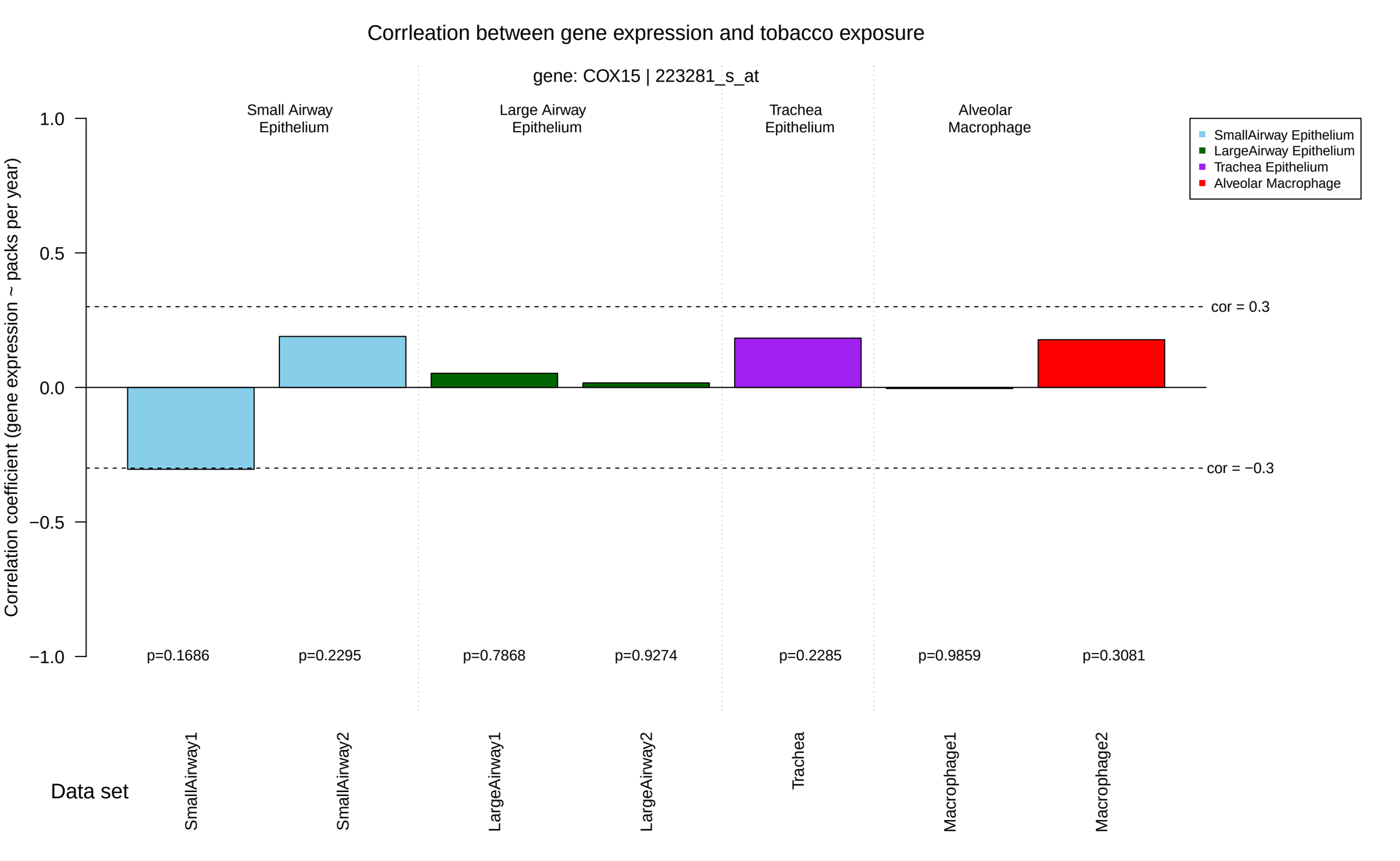

set: LargeAirway1 (p=0.2997)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7565)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0468)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7743)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3733)
COX15 | 223281_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX15 | 223281_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1686
p=0.2295
p=0.7868
p=0.9274
p=0.2285
p=0.9859
p=0.3081
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2